



Data Intelligence Protocol

DATA INTELLIGENCE PROTOCOL INC.

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FINAL SECURITY ASSESSMENT

MapleFiToken Smart Contract

Comprehensive Source-Code Security Audit

Technical assessment of the fixed-supply MapleFi (MPFI) ERC-20 implementation, covering supply integrity, access control, pause behavior, EIP-2612 permit handling, rescue functions, delayed default-admin governance, compilation, static-analysis triage and executable security testing.

OVERALL SECURITY SCORE

95 / 100

Internal weighted assessment indicator

ASSESSMENT OPINION

PASS WITH OPERATIONAL CONDITIONS

No Critical or High severity source-code vulnerability was identified. Residual risk relates primarily to privileged-role custody, treasury concentration and live-deployment attestation.

0

CRITICAL

0

HIGH

2

MEDIUM

2

LOW

2

INFORMATIONAL

BASE MAINNET DEPLOYMENT - RECORDED CONTRACT ADDRESS

0x43DCd925BfC9844623Ca9356D15D8F97AD566173

Recorded for identification and operational verification. Source-code approval does not by itself establish deployed-bytecode parity; the verification checklist appears in Section 09.



00 Document Control

Report Identification

FIELD	DETAIL
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Prepared for	MapleFi Project
Prepared by	Data Intelligence Protocol Inc. Security Review Team
Engagement	Smart-contract source-code security assessment
Distribution	Public release - complete and unmodified
Classification	Public

Corporate Identity

FIELD	DETAIL
Legal name	DATA INTELLIGENCE PROTOCOL INC.
Jurisdiction	Alberta, Canada
Incorporated	May 6, 2025
Corporate identifiers	Access No. 2027103635 Business No. 762390367
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Source Identification and Validity

ITEM	RECORDED VALUE
Reviewed file	MapleFiToken.sol
Source SHA-256	d459b14694650ab7ec4662f3db70b90c273c11545b99ef6ad894a984fa51b99c
Compiler / dependency	Solidity 0.8.20 OpenZeppelin Contracts 5.0.0 import paths
Validity condition	This opinion applies to the reviewed source hash and to a deployed instance only when runtime bytecode and constructor parameters are demonstrated to match the reviewed build.

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01 Executive Summary

Data Intelligence Protocol Inc. reviewed the supplied MapleFiToken Solidity source as a fixed-supply, non-upgradeable ERC-20 token intended for deployment on Base. The implementation is concise and relies on OpenZeppelin Contracts 5.0.0 for standard token behavior, capped supply, EIP-2612 permit handling, global pause controls, delayed default-admin transfer and guarded recovery operations.

95

SECURITY SCORE

ASSESSMENT OUTCOME

Pass with operational conditions

No Critical or High severity source-code vulnerability was identified. The two Medium findings are governance and custody conditions rather than unrestricted code execution, arbitrary minting or user-balance seizure defects.

0

CRITICAL

0

HIGH

2

MEDIUM

2

LOW

2

INFORMATIONAL

Key Security Strengths

SUPPLY INTEGRITY

One-time fixed supply. The complete 1,000,000,000 MPFI supply is minted in the constructor; no post-deployment mint entry point is exposed.

Hard cap. ERC20Capped enforces the same maximum supply at the shared update path.

IMMUTABLE TOKEN LOGIC

No upgrade mechanism. No proxy, delegatecall, implementation setter, initializer or upgrade authority was identified.

Standard accounting. No transfer tax, blacklist, anti-sell rule or privileged seizure hook was identified.

ADMINISTRATIVE SAFEGUARDS

Delayed admin transfer. The default administrator is transferred through OpenZeppelin's two-step delayed process.

Bounded delay. Current and future admin-transfer delays are restricted to 1-30 days.

TRANSACTION SECURITY

Permit protection. EIP-712 domain separation, nonces, deadlines and ECDSA validation are inherited from OpenZeppelin.

Recovery protection. Role checks, SafeERC20, Address.sendValue and ReentrancyGuard protect rescue functions.

02 Scope of Review

In-Scope Components

COMPONENT	STATUS	ASSESSMENT COVERAGE
MapleFiToken.sol	FULL REVIEW	178 source lines covering constructor logic, ERC-20 operations, role controls, pause, permit, cap, burn, rescue, native-currency receipt, inheritance overrides, events and custom errors.
OpenZeppelin Contracts 5.0.0	DEPENDENCY REVIEW	ERC20, ERC20Burnable, ERC20Capped, ERC20Pausable, ERC20Permit, AccessControlDefaultAdminRules, SafeERC20, Address and ReentrancyGuard integration.
Local assessment build	COMPILED	Solidity 0.8.20; optimizer enabled at 200 runs; ABI and bytecode inspected.
Executable security tests	9 / 9 PASSED	Authorization, constructor validation, cap and supply, pause coverage, permit replay, rescue security, reentrancy, delayed administration, delay bounds and state invariants.

Security Questions Addressed

SUPPLY

Can supply be increased after deployment, bypass the cap or be modified through hidden accounting?

AUTHORITY

Can privileged roles mint, seize user balances, upgrade logic or transfer arbitrary holder assets?

AVAILABILITY

What operations are blocked by pause, who controls pause and what availability risk remains?

SIGNATURES

Does EIP-2612 permit enforce nonce, deadline, domain and signature validation requirements?

RECOVERY

Can recovery functions be re-entered or used outside assets held by the token contract?

ADMINISTRATION

Is default-admin transfer delayed and are current and future delay values bounded?

Out of Scope

Private keys and signer devices; multisig deployment and signer composition; treasury operations; vesting, distribution, launchpad and liquidity-lock contracts; DEX contracts; front-end applications; APIs and infrastructure; bridges; exchange integrations; market behavior; and legal, tax or regulatory compliance were not supplied as part of this source-code assessment.

Deployment limitation. The Base mainnet address in this report was supplied by the client. Runtime-bytecode parity, constructor arguments, current role holders, pause state and on-chain event history require independent explorer or RPC verification before the live address is represented as matching the reviewed source.

03 Methodology and Assumptions

Assessment Methodology

01 MANUAL REVIEW

Line-by-line inspection of authorization boundaries, state transitions, inheritance resolution, token economics, external calls, error handling and failure modes.

02 COMPILATION

Local compilation using the declared Solidity generation and exact OpenZeppelin import paths, followed by ABI and bytecode inspection.

03 STATIC ANALYSIS

Slither-assisted review with manual triage to separate contract-specific observations from dependency, compiler and non-applicable notices.

04 SECURITY TESTS

Targeted Hardhat tests for critical properties, negative authorization paths, permit replay, delayed administration and reentrancy resistance.

05 INVARIANT TESTING

150 deterministic randomized transfer and burn operations validating aggregate balances, cap preservation and supply reduction only through burns.

06 RISK CLASSIFICATION

Impact, exploitability, required privileges, user exposure, recoverability and operational context were considered for each observation.

Methodology References

The review process was informed by common smart-contract security practices and control categories described by the OWASP Smart Contract Security Verification Standard and the EEA EthTrust Security Levels specification. Reference to these frameworks does not constitute OWASP or EEA certification.

Core Assumptions

- **Dependency assumption.** Third-party library correctness is relied upon to the extent of the identified OpenZeppelin version and its observed integration. New dependency or compiler disclosures may change this assessment.
- **Point-in-time validity.** Conclusions apply to the supplied source hash and information available on the issue date. A code, dependency, build, deployment, role or configuration change requires reassessment.
- **Operational assumption.** Private-key custody, multisig policy, treasury controls, monitoring and incident response are implemented outside the token source and materially affect residual risk.
- **Economic limitation.** This assessment evaluates source-code security properties; it does not predict token price, liquidity, market conduct, exchange behavior or economic outcomes.

04 Contract Architecture and Authority Model

OpenZeppelin Components

- ERC20 / Burnable / Capped / Pausable
- ERC20Permit (EIP-2612)
- AccessControlDefaultAdminRules
- SafeERC20 / Address / ReentrancyGuard



MapleFiToken

- MapleFi (MPFI), 18 decimals
- 1,000,000,000 MPFI fixed cap
- One-time constructor mint to treasury
- Pause and recovery operations

Role and Authority Model

AUTHORITY	PERMITTED ACTIONS	SECURITY BOUNDARY
DEFAULT_ADMIN_ROLE	Grant and revoke operational roles; schedule, accept or otherwise manage bounded delayed default-admin changes through inherited rules.	Does not provide minting, arbitrary transfers, upgrades or direct user-balance access. Compromise can alter PAUSER_ROLE or RESCUER_ROLE membership immediately.
PAUSER_ROLE	Call pause() and unpause().	Can stop transfers, transferFrom operations and burns; cannot transfer or confiscate user tokens.
RESCUER_ROLE	Recover ERC-20 or native assets held by the token contract.	Cannot move assets from arbitrary holders or the treasury. Can move MPFI only when MPFI is held by the token contract itself.
Treasury	Receives the complete initial supply and may transfer or approve it under standard ERC-20 rules.	Custody is outside the token contract; no built-in vesting, allocation or distribution restrictions are enforced.

Authority Boundaries

NO POST-DEPLOYMENT MINT

No role or external function can increase MPFI supply after construction.

NO ARBITRARY SEIZURE

Administrative roles cannot debit balances from arbitrary token holders.

NO UPGRADE PATH

No proxy, implementation slot, delegatecall or upgrade authority was identified.

PAUSE IS AVAILABILITY CONTROL

Pause blocks movement and burns but does not grant custody over holder assets.

05 Security Assessment and Findings Summary

Security Property Assessment

CONTROL AREA	RATING	ASSESSMENT
Supply integrity	STRONG	Supply is created once, equals the cap at deployment and can only decrease through holder-authorized burns.
Standard transfer behavior	STRONG	No fee-on-transfer, blacklist, anti-sell, max-wallet, rebasing, privileged transfer or hidden balance adjustment was identified.
Permit and replay resistance	STRONG	EIP-712 domain separation, nonces, deadlines and ECDSA checks are inherited from OpenZeppelin ERC20Permit.
Emergency controls	GOOD	Pause is comprehensive and effective, but availability depends on disciplined role custody and incident governance.
Administrative transfer	STRONG	Default-admin transfer uses a delayed two-step process and remains bounded between 1 and 30 days.
External-call safety	STRONG	Recovery uses SafeERC20 and Address.sendValue; nonReentrant blocks recursive recovery attempts.
Upgrade and proxy exposure	NONE	No upgrade mechanism, proxy slot, delegatecall or implementation administration was identified.
Deployment evidence	VERIFY	Publish the exact build manifest and confirm deployed runtime-bytecode parity before representing the live address as matching this review.

Findings Register

ID	SEVERITY	TITLE	STATUS
MF-TKN-001	MEDIUM	Privileged role custody and role administration	OPEN
MF-TKN-002	MEDIUM	Full-supply treasury concentration	OPEN
MF-TKN-003	LOW	Global pause creates availability and centralization risk	DESIGN
MF-TKN-004	LOW	Compiler and build reproducibility controls	OPEN
MF-TKN-005	INFO	Rescue-role scope and operational disclosure	NOTED
MF-TKN-006	INFO	Live deployment attestation remains separate	VERIFY

Code-level conclusion: no hidden minting, unrestricted transfer authority, proxy upgrade path, transfer-tax logic, blacklist, arbitrary balance modification or reentrancy-based loss path was identified in the supplied source.

06 Privileged Role Custody and Administration

MF-TKN-001**MEDIUM****AFFECTED AREAS**

DEFAULT_ADMIN_ROLE, PAUSER_ROLE and RESCUER_ROLE; constructor lines 67-98; pause/unpause lines 104-113; rescue functions lines 119-145.

OBSERVATION

The default admin controls operational-role membership. OpenZeppelin's delayed two-step process protects transfer of the default-admin position, but individual grants or revocations of PAUSER_ROLE and RESCUER_ROLE are not timelocked. PAUSER_ROLE can suspend transfers and burns; RESCUER_ROLE can withdraw assets held by the token contract.

IMPACT

Compromise or misuse of a privileged key could create availability disruption, unauthorized recovery of assets held by the token contract or rapid reassignment of operational roles. The authority does not permit minting, upgrades or seizure of arbitrary holder balances.

LIKELIHOOD

Likelihood depends on production key custody, signer separation, wallet security, transaction-review practices and monitoring. Risk is materially higher if any privileged role is assigned to a single externally owned account.

RECOMMENDATION

Assign default-admin, pauser and rescuer authority to appropriately governed multisig wallets. Separate emergency and operational signers where practical; require hardware-backed keys; simulate transactions; monitor role, pause, admin and rescue events; and maintain documented approval, escalation and incident-response procedures.

STATUS

OPEN - OPERATIONAL. Validate current on-chain role holders, signer thresholds and role-separation policy before public reliance on the deployment.

07 Full-Supply Treasury Concentration

MF-TKN-002**MEDIUM****AFFECTED AREAS**

MAX_SUPPLY at line 44 and constructor mint at line 98.

OBSERVATION

The constructor mints the complete 1,000,000,000 MPFI supply to the designated treasury. The token contract intentionally contains no internal allocation schedule, vesting logic, transfer lock, distribution cap or treasury spending control.

IMPACT

A compromised or improperly governed treasury could transfer a substantial portion of supply, affecting distribution integrity, circulating supply, liquidity, market confidence and token-holder expectations. This is a custody and token-operations risk rather than a hidden minting vulnerability.

LIKELIHOOD

Likelihood depends on treasury wallet architecture and the use of transparent vesting, distribution, custody, launchpad, liquidity-lock and allocation contracts. Single-key or poorly governed custody materially increases exposure.

RECOMMENDATION

Use a secure multisig treasury with independent hardware-backed signers. Place time-bound allocations in verified vesting or distribution contracts; reconcile published tokenomics with on-chain balances; adopt transaction-review and change-control procedures; maintain treasury alerts; and publish material wallet designations for transparency.

STATUS

OPEN - OPERATIONAL. Confirm treasury custody, allocation controls and vesting/distribution arrangements independently of this token source.

08 Low and Informational Findings

MF-TKN-003 Global Pause Creates Availability Risk

LOW

OBSERVATION	ERC20Pausable is applied at the shared _update path, so pause blocks transfers, transferFrom operations and burns. The pauser may maintain the paused state without an on-chain maximum duration.
IMPACT	A compromised or misused pauser can interrupt exchange, liquidity, custody and user-transfer activity. Pause does not authorize confiscation or minting.
RECOMMENDATION	Use a multisig pauser, define objective triggers and response-time expectations, monitor pause events, maintain an incident runbook and publish material incidents.
STATUS	INTENDED DESIGN. Accept only with documented governance and monitoring.

MF-TKN-004 Compiler and Build Reproducibility Controls

LOW

OBSERVATION	The source pins Solidity 0.8.20 and OpenZeppelin 5.0.0 import paths. Deployed-code assurance nevertheless requires exact production build inputs and settings.
IMPACT	Incomplete build evidence can prevent third parties from proving that deployed or verified bytecode corresponds exactly to the audited source and dependency set.
RECOMMENDATION	Publish Standard JSON compiler input, solc commit, optimizer and EVM settings, metadata configuration, remappings, dependency lockfile, constructor arguments and bytecode hashes.
STATUS	OPEN. Complete as part of the public deployment manifest.

MF-TKN-005

INFO

Rescue-Role Scope and Disclosure

OBSERVATION

RESCUER_ROLE can transfer ERC-20 and native assets held by the token contract, including MPFI accidentally sent to the token contract address. It cannot move arbitrary holder or treasury balances.

IMPACT

The authority is limited but may be misunderstood without clear public disclosure.

RECOMMENDATION

Document the limited scope, use approved destinations, retain transaction records and monitor RescueERC20 and RescueNative events.

STATUS

NOTED. Operational transparency item.

MF-TKN-006

INFO

Live Deployment Attestation

OBSERVATION

The Base mainnet address was supplied by the client. The engagement package did not include an independently captured RPC response, verified Standard JSON build, constructor data or role-state attestation.

IMPACT

Source-code approval alone does not establish that the deployed instance matches the reviewed build or intended configuration.

RECOMMENDATION

Complete the checklist on page 11 and retain a timestamped evidence package. Distinguish source review from deployed-instance verification until parity is established.

STATUS

VERIFY. Deployment-evidence requirement.

09 Testing Evidence and Deployment Checklist

SECURITY TESTS

9 / 9

Targeted Hardhat cases passed.

INVARIANT OPERATIONS

150

Randomized transfers and burns.

LOCAL RUNTIME SIZE

9,749 B

Optimized assessment-build bytecode.

Executable Test Results

✓	Full fixed cap is minted once to the treasury and no mint function is exposed.	PASSED
✓	Zero constructor addresses and admin delays outside 1-30 days are rejected.	PASSED
✓	Only PAUSER_ROLE can pause; transfers and burns fail while paused and resume after unpause.	PASSED
✓	EIP-2612 permit succeeds for a valid signature and the same signature cannot be replayed.	PASSED
✓	Rescue functions enforce role checks, input validation and ERC-20/native recovery behavior.	PASSED
✓	Recursive native rescue is blocked even where the receiver holds RESCUER_ROLE.	PASSED
✓	Default-admin transfer requires the scheduled delay and recipient acceptance.	PASSED
✓	Post-deployment admin-delay changes remain bounded between 1 and 30 days.	PASSED
✓	Aggregate balances equal total supply and supply never exceeds the cap across 150 operations.	PASSED

RECORDED CONTRACT ADDRESS

0x43DCd925BfC9844623Ca9356D15D8F97AD566173

VERIFICATION ITEM	STATUS	REQUIRED EVIDENCE
Explorer source verification	CONFIRM	Verified source and exact compiler/build settings visible on the Base explorer.
Runtime-bytecode parity	CONFIRM	Timestamped comparison between on-chain runtime bytecode and the audited build.
Constructor parameters	CONFIRM	Treasury, initial admin, pauser, rescuer and adminTransferDelay decoded and retained.
Supply and cap	CONFIRM	cap(), totalSupply(), treasury balance and decimals read from the live contract.
Role holders	CONFIRM	defaultAdmin(), pendingDefaultAdmin(), PAUSER_ROLE and RESCUER_ROLE membership or event-derived membership.
Pause and admin delay	CONFIRM	paused(), defaultAdminDelay(), pending changes and recent administration events.
Operational monitoring	IMPLEMENT	Alerts for pause, role/admin changes, rescue activity and material treasury transfers.

10 Final Opinion and Professional Notices

Final opinion. Based on the reviewed source hash, manual review, local compilation, static-analysis triage and executable security tests, MapleFiToken is a well-scoped fixed-supply ERC-20 implementation with no identified Critical or High severity source-code vulnerability. It is suitable for its stated token purpose provided that privileged roles and treasury assets are securely governed and the live Base deployment is proven to match the reviewed build.

Legal and Professional Notices

Nature of the Engagement

The term "audit" is used in the smart-contract industry sense. This report is a technical security assessment, not a financial statement audit, accounting assurance engagement, legal opinion, regulatory approval, insurance policy or formal accreditation.

Scope and Point-in-Time Validity

Conclusions apply only to the identified source hash and information available on the issue date. A source, dependency, compiler, constructor, role, wallet, chain, deployment or configuration change may invalidate them.

No Guarantee of Security

Security testing reduces risk but cannot prove the absence of every defect, exploit, economic attack, compiler issue, dependency defect, operational failure, key compromise or future vulnerability.

Third-Party Systems

OpenZeppelin components, wallets, multisigs, launchpads, vesting contracts, DEX infrastructure, bridges, front ends, APIs, Base network behavior and other third-party systems are outside scope except where direct integration is described.

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